Raspberry Pi 4B + L298N wiring, on official photos

Left: Raspberry Pi official 4B photo. Right: SunFounder official L298N pin photo. The table is this project's wiring plan.

Raspberry Pi 4 Model B (official photo)



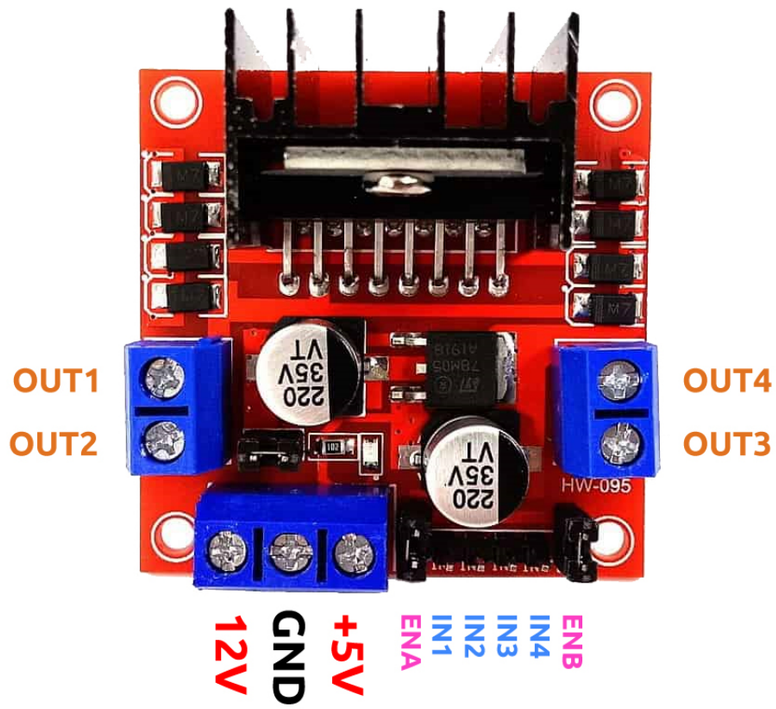
The 40-pin GPIO header is on the board edge. Find pin 1 first, then count.

Page 2 has the official enlarged pin map.

Wire by wire

Pi pin / function	-> destination
2 / 5V power	-> +5V (see red note)
6 / GND	-> GND
11 / GPIO17	-> IN1 right
13 / GPIO27	-> IN2 right
12 / GPIO18	-> ENA right
15 / GPIO22	-> IN3 left
16 / GPIO23	-> IN4 left
33 / GPIO13	-> ENB left
40 / GPIO21	-> Brush relay IN
32 / GPIO12	-> Buzzer

L298N: SunFounder official pin photo



Power everything off first, then remove the ENA, ENB and 5V-EN jumpers.

Only connect Pi pin 2 (5V) to the board's +5V after the 5V-EN regulator jumper is removed. Never put 5V on a GPIO pin.

Power

Pi: its own regulated 5V / 3A supply through USB-C.

Motor supply: + through a switch to the 12V/VS terminal, - to GND.

Pi GND, driver GND and motor-supply negative share one ground.

Check the motor voltage before switching on; never join the two positives.

Motors, brush and camera

OUT1 + OUT2: right-side motors. OUT3 + OUT4: left-side motors.

Two motors per side in parallel: check their total current first.

Camera ribbon -> Pi CAMERA (CSI), not the GPIO header.

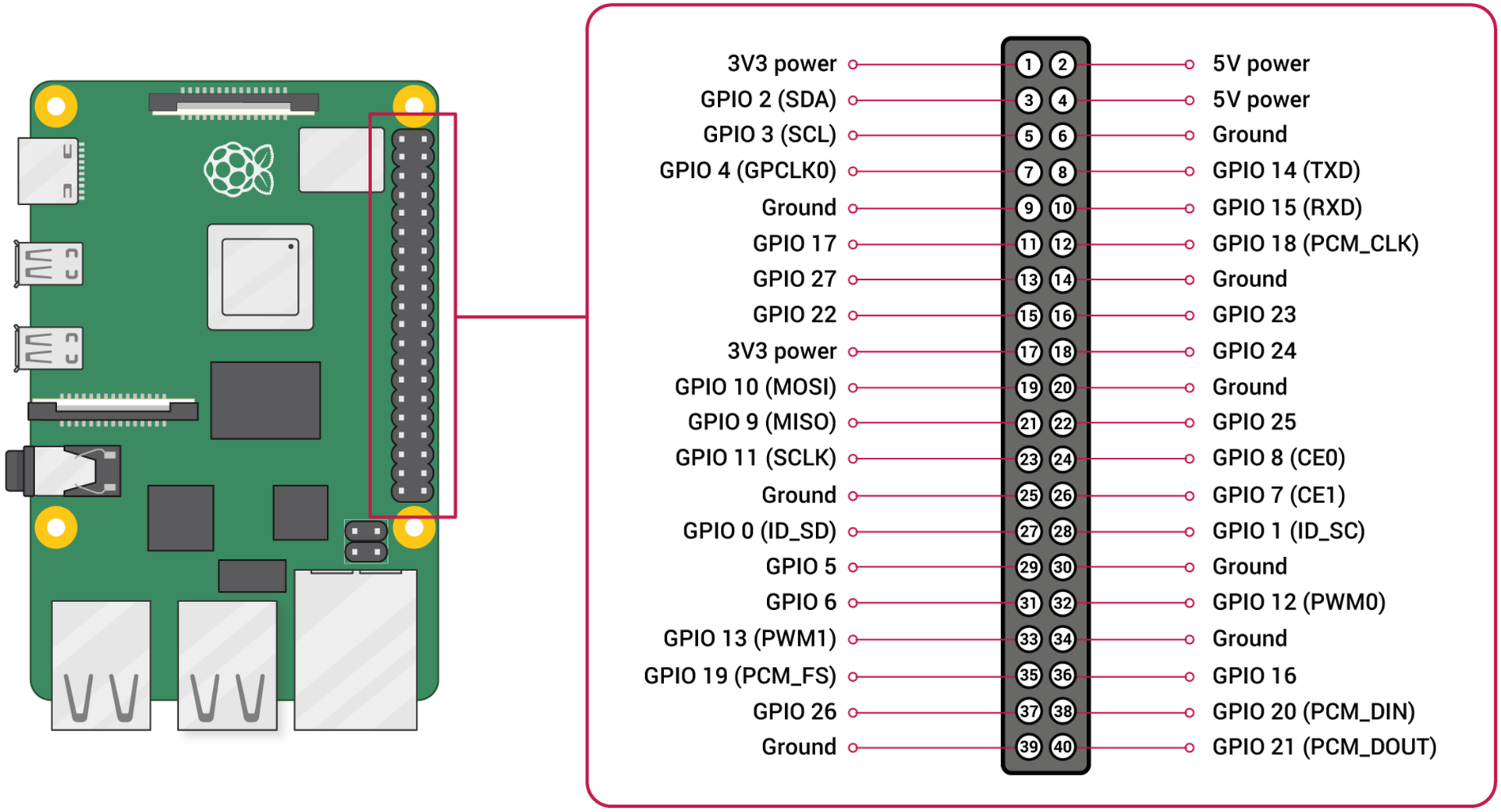
Brush: relay on GPIO21 (pin 40). First motor test with wheels in the air.

Raspberry Pi official documentation

SunFounder official documentation

Official Raspberry Pi 40-pin header (applies to the Pi 4B)

The board outline is generic; wire by the circled physical numbers 1-40. GPIO17 is physical pin 11.



Physical pins used by this project: 2, 6, 11, 12, 13, 15, 16, 32, 33, 40.

Software GPIO18 = jumper wire on physical pin 12; GPIO13 = pin 33; GPIO21 = pin 40; GPIO12 = pin 32.

"3V3 power" in the official image is the 3.3V supply, not 5V logic. Pin 2 (5V power) is not GPIO2.

Source: Raspberry Pi official hardware documentation / GPIO